

Bayesian Longitudinal Maize Height Estimation with Prior-Guided Image Analysis

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Abstract

Affordable longitudinal field phenotyping from fixed cameras is limited by ambiguous detections and uncertain pixel-to-centimeter calibration. Most pipelines analyze each image separately and apply a longitudinal model only after measurement extraction. We instead feed the Bayesian predictive state into current-image analysis: predicted position guides plant association, and the predicted height distribution can score competing top candidates before the measurement enters the longitudinal update. A robust particle filter updates calibrated plant height and growth. Each fixed-setting update uses images through time t and returns an as-of height distribution. Held-out pole bands evaluated calibration across cameras and years. Independent physical-height validation used 132 manual measurements from 33 plants across 12 stationary-camera rows in 2021, with manual height excluded from fitting and tuning. Mean absolute error decreased from 12.03 to 10.97 cm (within-record reduction 1.06 cm; 95% row-cluster bootstrap interval, -0.16 to 2.39). A Gaussian local-linear Kalman filter achieved 11.62 cm; its within-record difference from the particle filter was uncertain (0.66 cm; -0.43 to 1.79). Posterior 95% intervals covered 94.7% of all 132 held-out manual heights (75.4-cm mean width); year-forward 2025 predictive coverage was 94.9% using 2024-derived parameters. In controlled ambiguity, prior-guided candidate scoring reduced latent-height root mean squared error from 10.5 to 6.0 cm, compared with 8.6 cm when filtering began after candidate selection; a matched real-image audit found no candidate conflict. The study supports affordable longitudinal phenotyping while locating the methodological innovation at the image-analysis decision.

Keywords: Bayesian longitudinal model; prior-guided image analysis; maize height; particle filter; stationary camera; uncertainty calibration

1 Introduction

Longitudinal plant-height trajectories describe growth rate, developmental timing, and response to environment more directly than a single mature measurement. Repeated manual measurement is costly and can disturb field plots, while fixed red-green-blue cameras can collect the same view throughout a season at comparatively low cost. Existing time-series phenotyping systems have produced useful growth curves in maize and other crops [1–3]. The practical obstacle is that a camera produces pixels rather than a physical measurement. Canopy overlap, occlusion, wind, changes in illumination, and the transition from leaf tips to reproductive structures make the top and base of an individual plant ambiguous. A separate reference object can establish scale, but its geometry and usable range must be known. In a conventional two-stage pipeline, each image is reduced to a fixed measurement before the longitudinal model sees it; the downstream model cannot use earlier images to resolve the current image-level choice. A longitudinal method must also state exactly when information becomes available and how uncertainty changes as new images accumulate.

Several parts of this problem have been addressed separately. Time-lapse cameras with an in-scene scale can measure crop height, including operator-assisted rice measurements [4, 5]. Stereo and structured imaging can estimate canopy height accurately, but require additional hardware or controlled acquisition [6, 7]. Fixed-camera maize methods have pooled information across images, fitted nondecreasing growth curves, and used early images as labels for later overlapping plants [8, 9]. Other field-maize pipelines reuse locations from earlier growth stages as seed points for later segmentation or use prior-stage maize positions to guide current time-series point-cloud segmentation and height extraction [10, 11]. These studies establish that temporal position information can guide later image analysis; they do not evaluate the same coupling of causal image analysis, physical calibration, and height-distribution updating studied here.

State-space models provide a formal way to carry information through time. Kalman and particle filters are standard tools for sequential estimation [12–14]. Particle filtering has been used for real-time rice-height estimation from Sentinel-1 images, and a recent Bayesian pattern-matching framework estimates daily rice height from satellite vegetation indices and sparse field measurements [15, 16]. A Kalman tracker has also been combined with a maize seedling detector for counting [17], while Bayesian adaptive sampling has been applied to affordable time-lapse germination phenotyping [18]. Bayesian inference necessarily uses prior information, so the mere presence of a prior is not our novelty claim. The distinction is architectural: the predictive state from earlier images is fed back into analysis of the current image before its measurement is finalized. In the field implementation, the running plant-position state guides association and base-position refinement. The stronger branch lets the predicted height distribution choose among competing top candidates; it is evaluated in controlled simulation and a matched real-image audit. Thus *prior-guided* modifies image analysis, while *Bayesian longitudinal* describes the height-growth method that supplies and updates the predictive state.

The study integrates three elements for longitudinal field phenotyping. First, prior-guided image analysis uses the predictive longitudinal state during current-image association and candidate selection rather than only after a measurement has been chosen. Second, in-scene physical references and recorded camera geometry convert the selected image extent to centimeters, with separate definitions for 2021 camera-support poles and 2024–2025 red-band calibration poles. Third, sequential, outlier-robust Bayesian updating returns a plant-specific height-growth distribution and exact-mixture predictive intervals. Online operation is an as-of update: when a new image is appended, the algorithm analyzes the available image prefix and reports a measurement after processing that image; later observations join the pool only when they arrive. The primary independent physical-height validation uses all available 2021 manual measurements across 33 plants, 12 stationary-camera rows, and five dates. Separate held-out tests address detector agreement, physical calibration, candidate ambiguity, and temporal transfer of predictive uncertainty.

2 Materials and Methods

2.1 Experimental and technical design

Figure 1 shows the information flow. Let $\mathcal{I}_{\leq t}$ be the images available through acquisition time t . The image arriving at t is appended to this pool, and the predictive state from $t - 1$ returns to image analysis before the measurement is fixed. The running position state guides association; in the stronger branch, predicted height scores top candidates. Selected top and base rows are converted to centimeters, incorporated into the Bayesian update, and returned as an as-of distribution. Images acquired after t are absent. The 2021 study independently validates this prefix-causal model: test plants were separate, all manual labels were held out from tuning, and the five reference dates ended with the end-of-season measurement.

The linked data sets have distinct roles (Table 1). Primary independent validation uses 2021 manual measurements on five dates and archived segmentation extents for all matched plant-date records across 12 stationary-camera rows; the final reference is at season end. Test plants are separate from 2024–2025 development data, and manual heights only score predictions. The 2024 development and 2025 locked-test pole images evaluate calibration, and a quality-controlled subset supplies the secondary prequential stream. Annual replanting prevents subject overlap across years, although repeated images within plants and cameras remain dependent. Separate 2021 human-annotated images support out-of-fold detector validation. Haoming Wang’s newly completed support-pole annotations describe 2021 images and audit installation geometry; they are not red-band calibration truth.

2.2 Physical references and geometry

Haoming Wang manually annotated the 2021 physical references in CVAT. The XML files contain labels `pole1` through `pole4`; each label identifies a different stationary-camera support pole visible in that image, and the number does not encode depth. The annotated vertical segment runs from ground contact to the nominal camera mounting or optical-center height, 5 ft (152.4 cm). The

Bayesian longitudinal estimation with prior-guided image analysis

As-of update at acquisition t : the result uses $I_{\leq t}$; I_{t+1} does not yet exist.

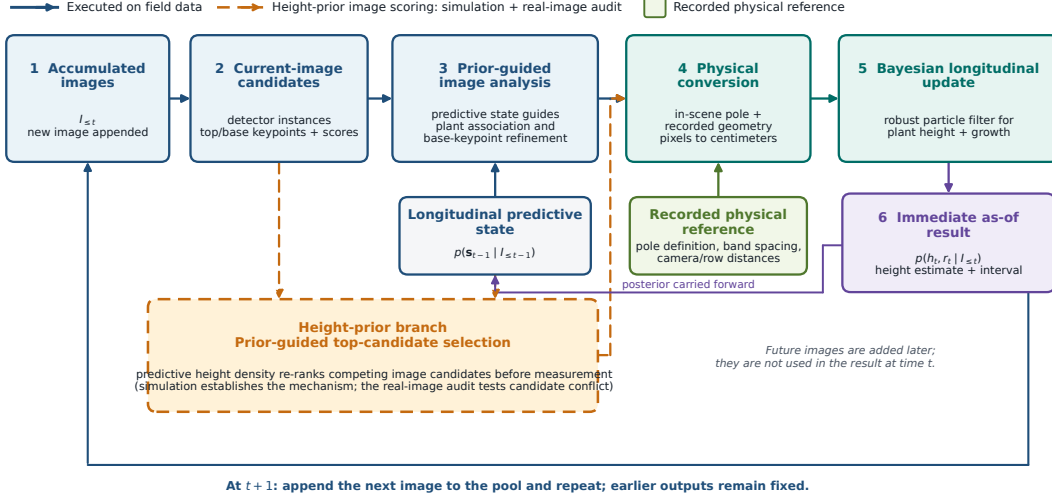


Figure 1: Workflow for the as-of image update. The feedback arrows place prior information inside current-image analysis: the running position state guides detection association and base refinement, and the dashed amber branch uses the predicted height distribution to choose among alternative top candidates. Solid blue stages were executed in the field pipeline; the height-candidate branch was evaluated in controlled simulation and the matched real-image audit. At time t , only $I_{\leq t}$ is available; with settings fixed, a later image starts a new update and does not alter an earlier output.

Table 1: Data sources, physical references, and inferential roles. Counts are reported at the unit used in each analysis.

Data set	Units	Physical or annotation reference	Role
2021 field heights	132 records; 33 plants; 12 stationary-camera rows; 5 dates	Manual height and geometrically scaled image extent	Primary independent manual-reference physical-height validation
2021 detector set	30 images; 150 masks	Highest and lowest pixels of each human mask	Five-fold out-of-image detection agreement
2021 support poles	12 XML files; 61 retained images; 199 human traces	Nominal 5-ft support pole; ground contact to camera mount/optical-center height	Human physical-reference and installation audit
2024 pole development	7 installed cameras; 6 usable in held-out validation	Red bands spaced 1 ft edge-to-edge	Calibration development and noise estimation
2025 locked test	3 installed and usable calibration cameras	Same red-band definition	Locked cross-camera/year calibration test
Quality-controlled sequential stream	1686 records; 48 tracks; 8 cameras; 330 images	Red-band calibration and camera geometry	Secondary next-image prediction and interval transfer
Candidate-ambiguity simulation	480 plants in 8 scenarios	Known latent height and correct candidate	Mechanism comparison with plant-cluster inference

field records also contain four horizontal layout intervals of 35 in, totaling 140 in between adjacent rows. These horizontal marks are not tick marks on the support poles. The confirmed duplicate C-039_2021-07-30.JPG .JPG was removed while C-039_2021-07-30 .JPG was retained. The completed human audit contains 199 traces on 61 unique XML-referenced images: 172 marked usable and 27 marked unusable (Figure 2).

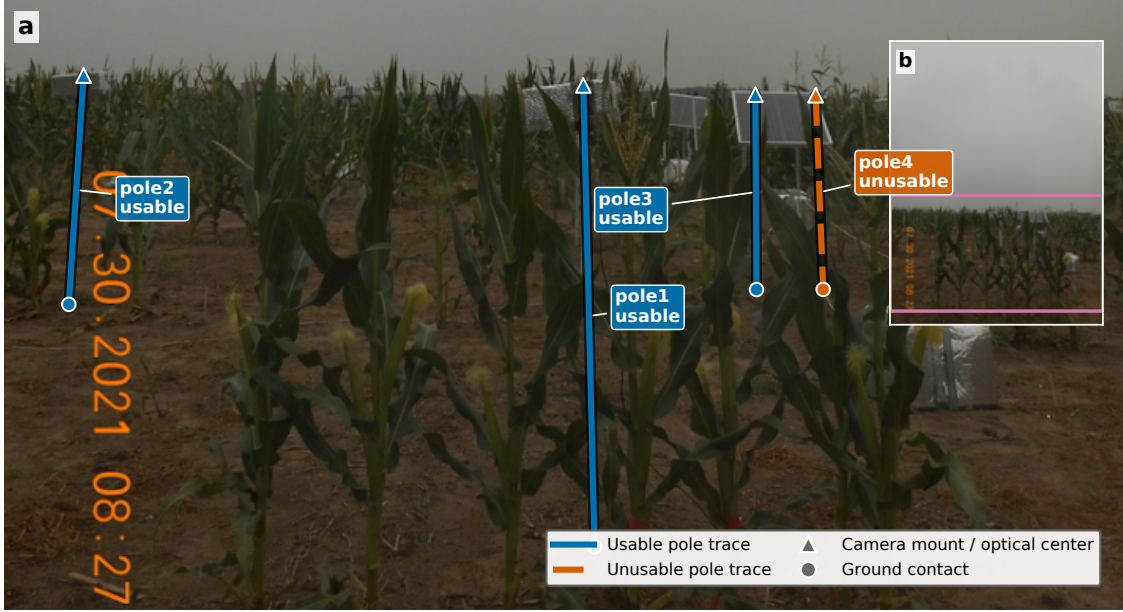


Figure 2: Representative 2021 camera-support-pole annotations in C-039_2021-07-30.JPG. Blue traces were marked usable and the orange trace unusable in the source XML. Circles denote ground contact and triangles the nominal camera mounting or optical-center height. Each traced support pole has a documented physical length of 152.4 cm. The inset shows the full retained image and crop boundary. The 35-in field marks recorded for this installation describe horizontal row layout and are not vertical pole graduations.

The 2024–2025 reference is a different object: an approximately 8–10-ft vertical calibration pole with red bands separated by 1 ft (30.48 cm), measured edge-to-edge. Its visible bottom is not assumed to be ground. Candidate band rows y_j are assigned relative heights z_j , and a projective map is fit within each accepted reference image,

$$z(y) = \frac{au + b}{cu + 1}, \quad u = \frac{y - \bar{y}}{s_y}. \quad (1)$$

Fits are rejected for insufficient bands, excessive line or height residuals, irregular spacing, nonmonotonicity, or a near-singular denominator. Because plant height is the difference between the mapped top and base rows, the arbitrary height datum cancels.

Recorded 2024–2025 geometry gives a camera height of 5.0 ft, camera-to-plant-row distance $D_{\text{tgt}} = 8.5$ ft, and camera-to-reference distance $D_{\text{ref}} = 10.0$ ft. Under a pinhole camera, negligible

roll, and a shared level ground plane,

$$y - y_{\text{horizon}} = f_{\text{px}} \frac{H - z}{D}. \quad (2)$$

Subtracting Equation 2 at the top and base shows that an extent calibrated at the pole depth is converted to the target depth by

$$h_{\text{target}} = \frac{D_{\text{tgt}}}{D_{\text{ref}}} h_{\text{pole}} = 0.85 h_{\text{pole}}. \quad (3)$$

Equation 3 is exact under the stated projection assumptions; uncertainty in distance, tilt, relief, lens distortion, wind, and maintenance remains empirical error. The 2021 single-frame height uses the separately documented camera geometry, 10.25 ft to the target row with a 4.5-mm focal length, 6.17-mm sensor width, and 1030-pixel stored field of view, giving 0.415886 cm/pixel with zero intercept.

2.3 Prior-guided image analysis within the Bayesian longitudinal method

For the 2024–2025 stream, a YOLOv8-pose model produces candidate plant instances with top and root keypoints. The first complete frame initializes six left-to-right tracks. At each later frame, observed root positions are matched to running track positions by minimum total horizontal distance, subject to a maximum normalized distance of 0.16. The track position and root row are then updated with a fixed gain of 0.5. This is prior-guided image analysis because the state carried from preceding images participates in current-image association and base refinement before the physical measurement is constructed. The sequential configuration associated 2018 detections and recovered 48 associations relative to frozen first-frame anchors; it changed the root used in the physical extent for 1963 matched records. The position recursion has the form of an exponentially weighted moving average, or a steady-state Kalman update with fixed gain. The deployed assembler does not estimate position covariance and is not a fully Bayesian neural network.

The 2021 all-plants comparison uses the archived per-image segmentation extent `height_sam`, multiplied by the recorded geometric scale. Manual field height is used only for scoring. The separate detector validation derives two keypoints from each human mask: the uppermost plant pixel and the lowermost plant pixel. This is an operational mask-extent definition, not an independently annotated flag-leaf collar. Five image-level folds were formed; a YOLOv8n-pose model was trained for 100 epochs on the other folds and evaluated on images that did not contribute to its training.

The height-prior candidate component isolates the same architectural question at the plant-top decision. Each current image in the simulation supplied one correct candidate with Gaussian error and, with probability 0.30, a competing high-confidence candidate displaced 25–60 cm upward or downward. Three estimators saw identical candidate sets: the highest-confidence single-frame candidate; the same independently selected candidate stream followed by a particle filter; and prior-guided image selection, which maximized detector confidence times the longitudinal predictive height density before filtering. The comparison therefore changes where the prior acts, not whether the downstream model is Bayesian. A complementary real-image audit used the same cached YOLO

157 candidates, fixed geometric scale, association gates, and downstream particle filter for both temporal
 158 methods. Human-mask root positions initialized tracks only in the first available image, and manual
 159 field heights were used only for scoring. This audit comprised 63 plant-dates from 17 tracks and 6
 160 cameras.

161 2.4 Bayesian filtering of longitudinal height and growth

162 We use *filtering* in the state-space sense: after image t is processed, the target is $p(h_t, r_t \mid Y_{1:t})$, the
 163 current height-growth distribution conditional on measurements available through t . This is distinct
 164 from spatial image filtering or denoising and from retrospective smoothing, which would use later
 165 measurements $Y_{t+1:T}$ to revise an earlier state.

For a plant observed at irregular interval Δ_t , the latent height h_t and growth rate r_t follow

$$h_t = h_{t-1} + \Delta_t r_{t-1} + \eta_t, \quad \eta_t \sim N(0, \sigma_h^2 \Delta_t), \quad (4)$$

$$r_t = r_{t-1} + \zeta_t, \quad \zeta_t \sim N(0, \sigma_r^2 \Delta_t), \quad (5)$$

166 with h_t truncated to $[0, 450]$ cm and r_t to $[-4, 15]$ cm/day. The lower growth bound permits
 167 short apparent decreases rather than forcing an irreversible monotone ratchet. The image-derived
 168 measurement Y_t has a robust two-component likelihood,

$$p(Y_t \mid h_t) = (1 - \pi) \phi(Y_t; h_t, \sigma_t^2) + \pi \phi(Y_t; h_t, \sigma_{\text{out}}^2), \quad (6)$$

169 where σ_t combines a base pose term, pole-fit residual, out-of-range distance, and band-regularity
 170 term. A bootstrap particle filter with 5000 particles propagates, weights, and resamples the state
 171 when effective sample size falls below half the particle count [14].

172 Two interval targets are kept separate. A posterior interval summarizes uncertainty in latent
 173 height after image t has been incorporated. A one-step predictive interval is formed before Y_t is
 174 observed and targets the next image-derived measurement. If predicted height particles are $h_t^{(m)}$
 175 with weights w_m , its cumulative distribution is

$$F_t(y) = \sum_m w_m \left[(1 - \pi) \Phi\left(\frac{y - h_t^{(m)}}{\sigma_t}\right) + \pi \Phi\left(\frac{y - h_t^{(m)}}{\sigma_{\text{out}}}\right) \right]. \quad (7)$$

176 The 0.025 and 0.975 quantiles are obtained by numerical inversion of Equation 7. This replaces the
 177 earlier Gaussian variance approximation, which did not preserve the outlier-mixture tails.

178 The daily 2024 development increments set $\sigma_h = 16.866$ cm/ $\sqrt{\text{day}}$, $\pi = 0.0512$, and $\sigma_{\text{out}} = 70.62$
 179 cm from a robust median/median-absolute-deviation core-tail split over 1172 within-track intervals.
 180 Height, measurement standard deviation, process scale, and outlier scale are all expressed after
 181 the same 0.85 pole-to-target depth conversion. These values combine process and measurement
 182 variation because one observation per plant-time cannot identify those components separately. A
 183 comparison run retains the prespecified guessed values 1.5 cm/ $\sqrt{\text{day}}$, 0.08, and 60 cm, also expressed
 184 in target-row units. No 2025 observation enters the empirical noise estimates, and no 2025 plant

subject appears in the 2024 development data.

For primary 2021 validation, per-image observation standard deviation ranges from 12 to 30 cm with wind, and process-height scale is estimated from image increments. A growth-rate field $\hat{r}(h)$ learned from 2024–2025 image increments is transferred to the subject-disjoint 2021 plants. No 2021 manual height enters the image measurement, growth field, filter, noise parameters, or comparator settings. Thus the 2021 data provide independent, cross-season, label-held-out physical-height validation. The external growth field introduces no 2021 label leakage, and every update remains prefix-causal. The separate 2024-to-2025 test evaluates year-forward predictive-interval transfer.

2.5 Validation and statistical analysis

Physical calibration was evaluated by fitting alternating 1-ft automatically detected red bands and predicting the withheld bands. Cameras, rather than band records, were the sampling units for bootstrap intervals. Six 2024 cameras formed the development set and three 2025 cameras were locked until the procedure was fixed. Root-direction extrapolation was evaluated separately: the lowest bands were removed, Equation 1 was fit only to the retained upper bands, and positions 1–3 ft below the fitted range were predicted. This direction matches plant-base use; the reverse lower-to-upper experiment is not described as root extrapolation. The completed human physical-reference audit comprises the separate 2021 support-pole annotations described above; because those support poles differ from the 2024–2025 red-band objects, they document installation geometry rather than serve as truth labels for the automatic red-band candidates. Temporal repeatability was summarized for every row–pole series with at least two usable traces by the coefficient of variation of annotated pixel length. Its median and 90th percentile were accompanied by 20,000 whole-row bootstrap replicates.

Detector agreement was summarized by detection rate, keypoint error, height-extent error, and correlation in five-fold cross-fitting. The candidate-ambiguity simulation used 60 plants in each of eight misspecified growth and observation scenarios. Mean plant-level root mean squared errors and their 95% intervals were calculated by 20,000 plant bootstrap replicates. The real-image candidate audit resampled whole cameras. Secondary predictive uncertainty was evaluated at 50%, 80%, 90%, and 95% levels, together with logarithmic score and weighted interval score; settings estimated from the earlier 2024 cohort were evaluated without retuning on the temporally later, subject-disjoint 2025 cohort. Forecast rows were not treated as independent biological subjects because measurements repeat within plant tracks and cameras. The primary independent manual-reference field analysis retained every available 2021 record and compared the robust particle filter with the single-frame extent, an exponentially weighted mean, a three-image running median, a fixed Holt level-trend smoother, and a Gaussian local-linear Kalman filter. Comparator settings were fixed without manual-height tuning. Within-record absolute-error differences were bootstrapped 20,000 times by resampling the 12 stationary-camera rows and preserving their internal dependence. A mechanism-aligned secondary analysis retained observations after each plant’s first image, when a preceding image was available to supply longitudinal information. Central 80% and 95% latent-height posterior intervals were compared with the held-out manual heights; coverage and mean width received the same whole-row bootstrap treatment. Endpoint and previously residual-screened analyses are reported

only as sensitivity analyses in the Supplementary Materials.

Prefix causality was tested by rerunning every camera-time prefix and comparing all values already emitted with the corresponding rows from the complete run. The audit covered 391 prefixes and 51,035 cumulative row comparisons.

2.6 Reproducibility

The public source package includes analysis scripts, fixed seeds, tabular outputs, all 61 curated 2021 pole-calibration images, 12 human-annotated XML pole files, the imported 199-trace audit table, the released 2021 pose checkpoint with its SHA-256 checksum and recovered embedded training configuration, a complete candidate cache for those images, figure sources, and manuscript source. The checkpoint records the software version, seed, deterministic setting, training arguments, and summary metrics; the referenced dataset file, exact training-image manifest, original training masks, and source commit were not recovered. Source-frame matching recovered coordinates and match scores for 87 legacy plant crops; 63 join the 132-record primary validation across six camera rows, leaving 69 primary extents without a reconstructable raw-image chain. Headline numbers are regenerated from canonical comma-separated-value or JavaScript Object Notation outputs rather than transcribed. The daily 2024–2025 raw-image archive remains too large for this release, so its published analyses start from checksum-identified derived measurements.

3 Results

3.1 Physical calibration and extrapolation

The 172 usable 2021 support-pole traces formed 39 repeated row–pole series across all 12 stationary-camera rows. Median within-series pixel-length coefficient of variation was 1.24% (95% whole-row bootstrap interval, 0.78–1.99%), and its 90th percentile was 3.65%. 38 of 39 series were below 5%; the remaining series reached 11.51%. This audit supports temporal stability of the projected references and their annotations; it does not establish absolute scale accuracy.

The alternating-band experiment produced a camera-cluster mean absolute error of 0.186 in (95% camera-bootstrap interval, 0.156–0.218) in 2024 development cameras and 0.208 in (0.158–0.262) in locked 2025 cameras (Table 2; Figure 3). The locked interval overlaps the development interval, which argues against a large year-specific failure but does not establish equivalence with only three test cameras.

Correctly directed root extrapolation showed little aggregate signed bias but increasing dispersion with distance below the fitted bands. At 1 ft below the retained range, 1110 predictions from 10 cameras had an unweighted camera-mean bias of 0.40 cm, mean absolute error of 1.91 cm (95% camera-bootstrap interval, 1.24–2.98), and root mean squared error of 3.95 cm. Across all 1–3-ft distances, unweighted camera-mean absolute error rose to 2.93 cm (1.86–4.64) and root mean squared error to 6.78 cm. One development camera contributed only five usable images and had much larger 3-ft error, visible in Figure 3; this is a failure condition rather than evidence for broadly reliable extrapolation.

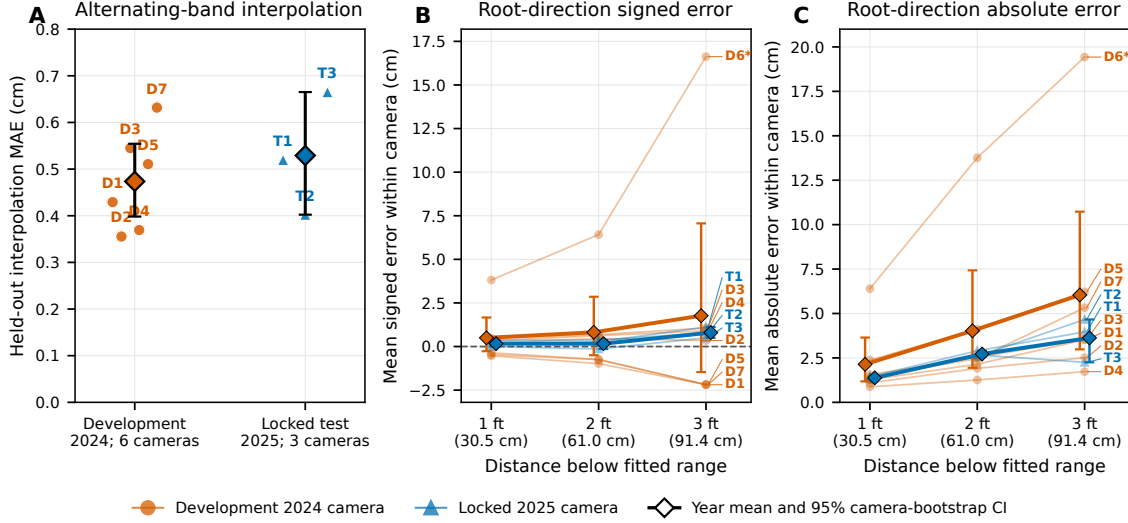


Figure 3: Camera-level physical-calibration validation. (a) Alternating-band interpolation mean absolute error for each 2024 development camera (D) and locked 2025 camera (T); diamonds and vertical intervals are unweighted year means and 95% whole-camera bootstrap intervals. (b) Mean signed error and (c) mean absolute error when lower bands are dropped, the retained upper bands are fit, and positions 1–3 ft below the fitted range are predicted. Thin lines are cameras; thick lines and intervals are unweighted camera means and whole-camera bootstrap intervals. D6, marked with an asterisk, has only five usable images. Individual bands are automatic candidates; the inferential unit is the camera.

Table 2: Held-out calibration performance. Interpolation intervals resample whole cameras. Root-direction rows give unweighted means of camera-level error summaries from the correctly directed upper-to-lower experiment.

Experiment	Cameras	Predictions	Bias	MAE	RMSE
Physical error					
2024 alternating bands	6	1083	–	0.186 in	–
2025 alternating bands	3	427	–	0.208 in	–
1 ft below fitted range	10	1110	0.40 cm	1.91 cm	3.95 cm
1–3 ft below fitted range	10	2162	0.58 cm	2.93 cm	6.78 cm

3.2 Prior-guided candidate selection

In the controlled ambiguity experiment, the single-frame rule had plant-level height root mean squared error 10.48 cm (95% plant-bootstrap interval, 9.84–11.13), and post-detection filtering reduced it to 8.61 cm (8.07–9.16). Allowing the predictive height distribution to score candidates before the update reduced it further to 6.01 cm (5.47–6.57; Figure 4). The within-plant reduction relative to post-detection filtering was 2.60 cm (95% interval, 2.00–3.20). Prior-guided selection chose the truth-closest candidate 97.0% of the time. Benefits varied by simulated condition and were smallest when images were missing or growth plateaued early. This experiment establishes the candidate-choice mechanism under known ambiguity; it is not a claim that the same effect size occurred in the field data.

In the matched real-image audit, post-detection filtering and height-prior candidate scoring selected the same candidate in 100.0% of 63 evaluated plant-dates. Their mean absolute errors were therefore identical at 13.07 cm, compared with 13.38 cm for the single-frame candidate. The natural-image subset contained no candidate-choice conflict under the prespecified gates, so this audit is a negative control rather than evidence for a field accuracy gain from candidate re-ranking.

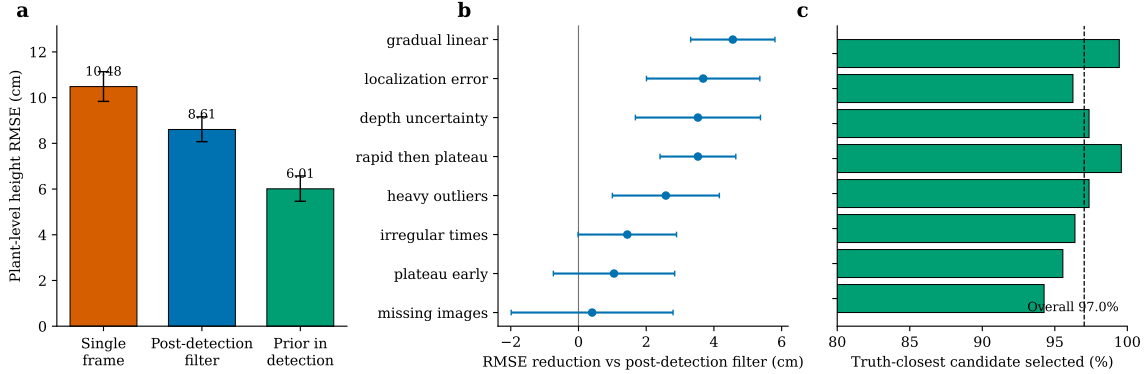


Figure 4: Controlled candidate-ambiguity experiment with 480 simulated plants across 8 scenarios. (a) Mean plant-level root mean squared error with 95% plant-bootstrap intervals. (b) Within-scenario reduction relative to post-detection filtering; points are means across 60 plants and bars are normal 95% intervals across plants. Positive values favor prior-guided candidate selection. (c) Percentage of frames in which prior-guided scoring selected the candidate closest to the known latent height. Every method saw the same candidates and used the same state dynamics; only the location at which the prior acted differed.

3.3 Independent physical-height validation

The primary independent validation used all 132 manual measurements obtained in 2021 from 33 plants across 12 stationary-camera rows and 5 dates. The test plants did not occur in the 2024–2025 development data, and manual heights were used only for evaluation; they did not enter the image measurement, growth field, filter, noise parameters, or comparator tuning. The single-frame image extent had signed error +2.64 cm, mean absolute error 12.03 cm, and root mean squared error 16.81

cm. The robust Bayesian posterior mean had corresponding values +2.73, 10.97, and 15.81 cm (Table 3; Figure 5). It had the lowest point-estimate error among the five fixed causal comparison methods. The within-record mean-absolute-error reduction relative to the single frame was 1.06 cm (95% row-cluster bootstrap interval, -0.16 to 2.39), and relative to the Gaussian local-linear Kalman filter it was 0.66 cm (-0.43 to 1.79). Both intervals include zero, so the observed advantage over the strongest comparators is modest and uncertain at this pilot scale. Against the three simpler longitudinal baselines, the corresponding gains were 4.01 cm (1.48–6.73) over EWMA, 7.80 cm (3.83–12.68) over the running median, and 3.08 cm (0.77–5.48) over Holt; all three row-cluster intervals excluded zero. The first image in a plant track cannot use preceding longitudinal information. In the mechanism-aligned secondary analysis of the 99 later records, mean absolute error decreased from 12.58 to 11.14 cm (gain, 1.44 cm; row-cluster interval, -0.21 to 3.26), retaining the direction of the all-record result with greater uncertainty.

Table 3: Independent physical-height validation in 2021. Every method is causal in image order and uses settings not tuned on manual height. The analysis includes all 132 plant-date records from 33 plants across 12 stationary-camera rows. Within-record method differences are evaluated by resampling whole camera rows.

Estimator	n	Signed error (cm)	MAE (cm)	RMSE (cm)
Single-frame image extent	132	+2.64	12.03	16.81
EWMA ($\alpha = 0.5$)	132	-9.78	14.98	20.16
Running median (3 images)	132	-12.15	18.77	27.14
Holt level-trend	132	-8.09	14.04	19.25
Gaussian local-linear Kalman	132	+2.74	11.62	16.52
Robust Bayesian particle filter	132	+2.73	10.97	15.81

3.4 Physical-height uncertainty and temporal transfer

Against the independent 2021 manual heights, central 80% latent-height posterior intervals contained 115 of 132 records (87.1%; 95% whole-row bootstrap interval, 79.3–93.6%) and averaged 47.9 cm wide (46.9–49.4 cm). Central 95% intervals contained 125 of 132 records (94.7%; 89.4–98.6%) and averaged 75.4 cm wide (73.5–78.3 cm). Manual height was excluded from interval construction. These are posterior agreement intervals after the current image, using manual height as the physical reference proxy; they do not estimate error in the manual measurement itself.

As a secondary temporal transfer check, exact mixture-based predictive evaluation materially changed the interpretation of uncertainty (Table 4). Guessed noise settings covered 88.6% of next image-derived measurements with a mean 95% width of 66.6 cm. After the depth correction was applied to every centimeter-scale noise quantity, the 2024-derived model covered 96.4% with a 98.1-cm mean width over the full stream. On the temporally later, subject-disjoint 2025 subset, coverage was 94.9% with a 97.8-cm mean width across 372 forecasts from 11 plant tracks and 2 cameras. Its weighted interval score was 8.60 cm. Plants were newly grown in 2025, so no biological subject recurred from the 2024 development cohort; repeated forecasts within each plant and camera remain dependent observations. Lower nominal levels remained conservative (Supplementary Figure S3), so

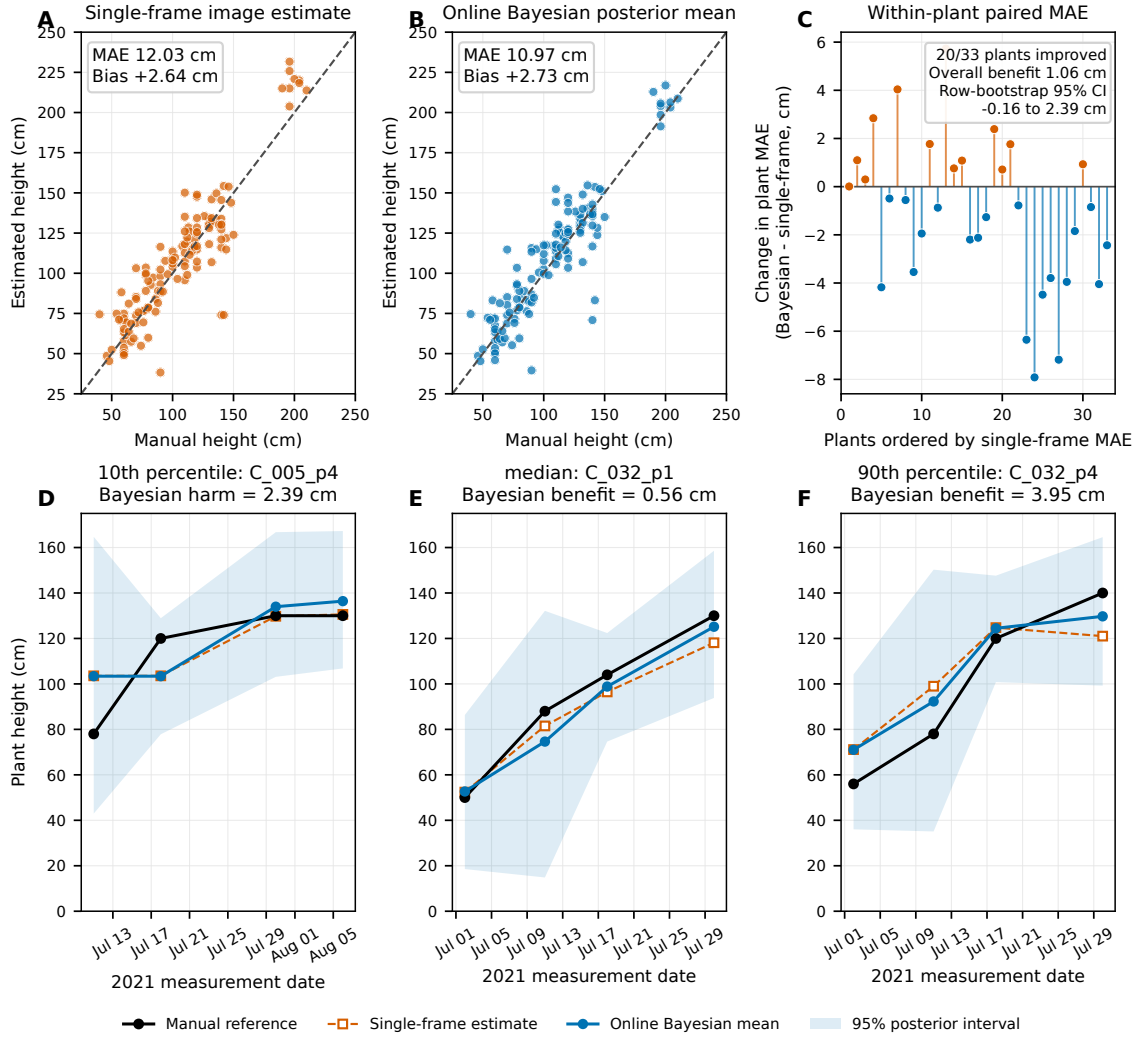


Figure 5: Independent 2021 manual-reference validation. (a,b) Single-frame and Bayesian estimates against manual height for all 132 records; dashed lines mark equality. (c) Within-plant difference in mean absolute error, ordered by single-frame error; negative values favor the Bayesian estimate. (d-f) Trajectories selected by the fixed 10th-, 50th-, and 90th-percentile rule on within-plant benefit, illustrating harm, median behavior, and benefit without residual-based exclusion. Shading is the 95% latent-height posterior interval. Within-record uncertainty is the stationary-camera-row bootstrap interval in panel c and Table 3.

the near-nominal 95% result should not be interpreted as complete distributional calibration. The calibrated Bayesian predictive mean had mean absolute error 15.13 cm and root mean squared error 21.34 cm, compared with 16.64/23.63 cm for the last observation carried forward and 19.95/27.52 cm for two-point linear extrapolation. This two-camera check assesses temporal interval transfer to the next noisy image measurement; it is not the primary physical-height validation or evidence of broad camera generalization.

Table 4: Uncertainty evaluations with distinct targets. Predictive intervals are formed before the next image-derived measurement; posterior intervals are formed after the current 2021 image and compared with manual field height.

Target	Configuration	n	Clusters	Coverage (%)	Mean width (cm)
Next image-derived measurement	Guessed noise, all years	1638	8	88.6	66.6
Next image-derived measurement	2024-derived, all years	1638	8	96.4	98.1
Next image-derived measurement	Secondary 2025 temporal test	372	2	94.9	97.8
2021 manual plant height	80% posterior agreement interval	132	12 rows	87.1	47.9
2021 manual plant height	95% posterior agreement interval	132	12 rows	94.7	75.4

3.5 Prefix-causality audit

All 391 camera-time prefix recomputations matched the corresponding rows of the complete run, with 0 mismatches among 51,035 cumulative row comparisons. This verifies prefix invariance for the cached detection-to-measurement pipeline under fixed settings. It does not show that model development occurred prospectively, and it does not cover raw-image detector training. Sequential position updating associated 2018 detections, compared with 1970 under frozen first-frame anchors, and reduced mean normalized association distance from 0.0396 to 0.0283. Because 2024–2025 manual plant heights are unavailable, these association results should be read as operational diagnostics rather than physical-height accuracy.

3.6 Human-annotated detector agreement

Five-fold detector validation matched 149 of 150 human masks (99.3%). The top keypoint was more difficult than the root (mean errors 28.3 and 18.2 pixels). Predicted vertical extent had an error of 29.5 pixels, corresponding under the recorded 2021 geometric scale to mean absolute error 12.27 cm, signed error +3.90 cm, and root mean squared error 15.91 cm; correlation with human mask extent was 0.837. These values replace the earlier centimeter conversion that had been fit using field height. They measure agreement with a human mask extent, not an independently clicked agronomic collar.

4 Discussion

This study develops a physically interpretable route from an accumulating longitudinal fixed-camera image stream to an as-of plant-height distribution after each image. The key methodological contribution is feedback across the usual image/model boundary: the predicted longitudinal state participates in current-image analysis before a measurement is committed. A running position state informs association and base refinement, the stronger candidate branch lets the predicted height distribution score alternative top detections, physical references convert the selected extent to centimeters, and a robust particle filter updates plant-specific height, growth, and uncertainty. Prior field-maize systems have reused earlier locations for later segmentation and height extraction [10, 11], and plant vision has used Kalman tracking [17]. Our contribution is the explicit prior-guided image-analysis loop inside a physically calibrated Bayesian longitudinal method, with separate evidence for each executed branch.

The evidence is strongest for component behavior and weakest for a large field accuracy gain. Repeated 2021 support-pole traces had a 1.24% median within-series coefficient of variation, supporting temporal stability of the projected reference. Held-out red-band interpolation transferred from six development cameras to three locked cameras with sub-inch error. The root-direction experiment also revealed why interpolation accuracy cannot be extrapolated casually: errors widened rapidly as the predicted point moved below the fitted band range, and one camera was particularly unstable. The candidate simulation showed that a height prior can prevent a confident but implausible candidate from entering the filter. The real-image audit, however, produced identical candidate choices for the two temporal methods. The height-prior candidate mechanism therefore remains demonstrated under controlled ambiguity rather than confirmed as a field advantage.

The primary physical-height evidence is the independent 2021 validation across 12 stationary-camera rows, 33 plants, and five dates. Test plants were absent from the 2024–2025 development data. Manual height was excluded from fitting and tuning, every matched record was retained, and the bootstrap resampled whole camera rows to preserve dependence. Mean absolute error decreased by about 1 cm relative to a single frame and by 0.66 cm relative to a Gaussian local-linear Kalman filter, but both camera-row-cluster intervals included no improvement. In contrast, intervals for the improvements over EWMA, running median, and Holt excluded zero. The mechanism-aligned after-first-image analysis retained the direction of improvement but remained imprecise. The 95% posterior interval covered 94.7% of the held-out manual heights, although its 75.4-cm mean width shows substantial uncertainty. The results do not support a claim that longitudinal Bayesian filtering uniformly improves height. They show that the robust particle filter achieved the best point estimate across the fixed causal baselines and supplied uncertainty with near-nominal 95% agreement without manual-height tuning.

The secondary temporal uncertainty check answers a different question. Applying the depth conversion to the observation, process, and outlier scales corrected a consequential unit mismatch. With parameters estimated from 2024 only, 95% coverage was 94.9% on the temporally later, subject-disjoint 2025 cohort with a 97.8-cm mean width, while 50%, 80%, and 90% intervals overcovered. Because the test contains only two cameras, it supports year-forward interval transfer

for those installations rather than broad cross-camera calibration. Repeated same-time images, direct uncertainty labels, or a hierarchical design are needed to separate measurement and biological variation and narrow intervals without losing coverage.

Several limitations determine the next experiments. First, the primary independent validation spans 12 stationary-camera rows but only 33 plants in one 2021 season. The secondary 2025 interval test has 11 plant tracks at two cameras; the locked calibration test has three cameras. Second, the 2024–2025 stream lacks manual plant-height outcomes. The independent 2021 study supplies physical-height truth on five dates through season end, using archived segmentation extents and an external growth field estimated without 2021 labels. Another subject-disjoint manual-reference season would strengthen transport evidence, but is not required for present independence. Third, human validation serves component-specific purposes: 150 plant masks assess image extent, 132 field measurements assess physical height, and Haoming Wang’s 199 newly completed traces of 61 2021 images audit physical-reference geometry. These sources do not estimate inter-reader variation for agronomic landmarks. Fourth, the depth ratio assumes known distances, level ground, negligible roll, and stable camera geometry. Fifth, the released checkpoint permits inference on the curated images, but its complete original training provenance remains only partially documented. Sixth, source-frame matching recovered positions for 87 archived plant crops; 63 of these map to the primary validation across six camera rows. For the remaining 69 primary records, the archive preserves the derived segmentation extent but not a reconstructable raw-image-to-extent chain.

A publication-scale extension could freeze all settings; evaluate without retuning in additional newly planted, subject-disjoint cohorts with manual reference heights; compare prior-in-image selection with post-hoc filtering using the same detector; and report cluster intervals matching the sampling design. The present 2021 study already supplies independent physical-height validation, and the 2024-to-2025 split validates temporal transfer for calibration and next-image prediction. An extension would add sample size, camera diversity, and external transport evidence.

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General

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Author Contributions

Haoming Wang, Chong Wang, and Peng Liu led conceptualization, methodology, software development, formal analysis, simulation, visualization, and writing of the original draft. Haoming Wang performed the physical-reference annotation and curated the associated image/XML audit. Yawei Li conducted the field work and curated field records. Patrick S. Schnable led experimental design, secured funding, provided resources, and supervised the field experiment. Cheng-Ting Yeh contributed validation. All authors contributed to validation, reviewed and revised the manuscript, and approved the final version.

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Conflicts of Interest

The authors declare that there is no conflict of interest regarding the publication of this article.

Data Availability

Analysis code, the 61 curated 2021 images, 12 human-annotated support-pole XML files, the imported 199-trace audit table, the released pose checkpoint, candidate predictions, derived data, figure sources, and manuscript source are available in the public GitHub repository. The repository also provides the 87-record partial source-frame coordinate recovery described above. It includes an MIT license for code and a SHA-256 manifest; no separate reuse license has been assigned to the non-code materials. The larger 2024–2025 raw-image archive remains in institutional storage; checksum-identified derived measurements needed for the reported analyses are public.

Declaration of Generative AI Use

Generative AI tools (Claude, Anthropic, and Codex, OpenAI) assisted data exploration, code review, figure and manuscript preparation, and language editing under direct author supervision. All numerical results were produced by the included analysis code, and the authors take responsibility for the design, verification, interpretation, and final text.

Supplementary Materials

Materials and Methods S1. Detailed model, causal-update, calibration, and reproducibility specification. Figure S1. Pinhole geometry and reference-to-target depth correction. Figure S2. Example posterior trajectories. Figure S3. Multi-level predictive-interval calibration. Table S1. Full parameter and implementation specification. Table S2. Predictive-interval calibration. Table S3. Real-image candidate audit. Table S4. Per-plant mature-date field comparison. Table S5. Data and result provenance. Data S1. Curated 2021 images and human support-pole annotations. Data S2. Canonical derived outputs and audit files. A separate Supplementary Materials PDF accompanies this manuscript.

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